

Introduction to Calculations of Heat loss on piping with Solar radiation

Przemyslaw Wiecek

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1 Introduction

At some point during one of the project I was a part of, it was necessary to check if the chosen thickness of insulation for long CIP water loops was enough, to keep the temperature control and over certain threshold. It turned out that one of calculations were done in a rush, resulting in wrong decision of making insulation thicker. Just in time, one of senior engineers stepped in and stopped the company from making somewhat pricy mistake.

This senior engineer, later send us the calculaitons to check it, write down things that were wrong and report back to him, to have a conversation about our findings. This way we could learn to spot errors. Because I like calculations and love learning about things inherently related to mathematics it was a match made in heaven. At this point in time I have not started my introductory book to mechanics of heat transfer by great J. Cengel, so it wasn't something I could jump right into. Thus, I took my time, and in around 2 weeks felt somewhat comfortable, at least with concepts (side note: I was especially taken by transient analysis and eigendecomposition of differential equations in plug flow problems, amazing, it might be something to write about) in steady state analysis to begin deriving equations and testing my knowledge.

The original calculations were made using iterative method, which is fine if the iterations are "fairly" small or the change in the underlying function is relatively insignificant or, even better, constant.

Funny note: I'm writting that I felt comfortable in around 2 weeks, but it took me around 5-6 months to understand that my first solution was completely wrong mathematically. Additionally, I had to rewrite my whole code because I convinced myself that all-functions approach is the way to go, which made my code unusable. Then I had to rewrite it again because of abovementioned mathematical error. All in all it took me around 9 months to have working template with this written as well.

2 Notation

Fluid and surface temperature are a function of length only, even if diameter changes or insulation thickness from one segment to another, those changes are also dependent on length. Thus, fluid and surface temperature will be written as $T_i(x)$, but when I will take a difference then length dependence is ommitted, for

example $T_f(x) - T_a = \Delta T_{fa}$. The value of K depends on single piping segment only on the value of h , which, because of used correlations, depends only on T_s , which will be always explicitly stated. Of course, as mentioned above surface temperature is a function of length, but in K and h this will be omitted for simplicity, and I just think it is easier to think of them as being a function of surface temperature only and not add length dependency. The only place it is stated is in chapter 3.2 in the first line for $\dot{Q}_{avg}$ for completeness, but then I drop it for the rest of this blog.

3 Derivation

3.1 Basics of steady-state heat transfer

The assumptions we must consider to make calculations possible are:

1. Steady-state process.
2. The change in kinetic and potential energy is negligible.
3. The flow is turbulent and pipes are full of liquid.
4. The fluid is incompressible.
5. The specific heat capacity is constant.
6. Fluid is losing energy to surroundings.
7. No axial heat transfer.
8. Uniform heat transfer distribution in piping cross section.
9. Piping is horizontal and enclosed by air only at ambient pressure and temperature.
10. Internal fluid heat transfer coeff. is infinite and material conductivities are constant.
11. Fluid is gaining energy by solar radiation (solar mode only).

With above in mind, one can utilize "Thermal Resistance Network" methodology and write that the average heat loss rate through a pipe segment of length L and distance x from the inlet is:

$$\dot{Q}_{avg}(x) = \frac{(T_f(x) - T_a)_{avg}}{R_{fa}} = 2\pi L K (T_f(x) - T_a)_{avg} \quad (1)$$

The *avg* notation means that we should take average of the temperature difference on piping segment of length L . With the same assumptions we can write that energy balance at any point must satisfy:

$$A_t \dot{q} - \dot{Q} = \dot{m} C d T_f \quad (2)$$

where:

- $T_f(x)$ - temperature of liquid at distance x , (F°, C°)
- T_a - ambient temperature, (F°, C°)
- A_t - total piping area (ft², m²)
- $\dot{q}$ - solar radiation flux ($\frac{\text{BTU}}{\text{ft}^2 \cdot \text{h}}$, $\frac{\text{W}}{\text{m}^2}$)
- R_{fa} - thermal resistance of piping of length from fluid to air L , ($\frac{\text{F}^\circ}{\text{BTU}}$, $\frac{\text{C}^\circ}{\text{W}}$)
- K - combined thermal conductivity, ($\frac{\text{BTU}}{\text{ft} \cdot \text{h} \cdot \text{F}^\circ}$, $\frac{\text{W}}{\text{m} \cdot \text{K}}$)

- $\dot{m}$ - mass flow rate, $\left(\frac{\text{lb}}{\text{h}}, \frac{\text{kg}}{\text{s}}\right)$
- C - specific heat capacity, $\left(\frac{\text{BTU}}{\text{lb}\cdot\text{R}}, \frac{\text{kJ}}{\text{kg}\cdot\text{K}}\right)$

and we define K as:

$$\frac{1}{K} = \frac{2N}{h_f D_i} + \frac{1}{k_{pipe}} \ln\left(\frac{D_o}{D_i}\right) + \frac{1}{k_{ins}} \ln\left(\frac{D_o + 2y}{D_o}\right) + \frac{2N}{h(D_o + 2y)} \quad (3)$$

where:

- y - insulation thickness, inch
- D_i - inner diameter, inch
- D_o - outer diameter, inch
- h_f - heat transfer coefficient from fluid to piping wall, $\left(\frac{\text{BTU}}{\text{ft}^2\cdot\text{h}\cdot\text{F}^\circ}, \frac{\text{W}}{\text{m}^2\cdot\text{K}}\right)$
- h - combined heat transfer coefficient from surface of a piping to air, $\left(\frac{\text{BTU}}{\text{ft}^2\cdot\text{h}\cdot\text{F}^\circ}, \frac{\text{W}}{\text{m}^2\cdot\text{K}}\right)$
- k_{pipe} - thermal conductivity of a piping, $\left(\frac{\text{BTU}}{\text{ft}\cdot\text{h}\cdot\text{F}^\circ}, \frac{\text{W}}{\text{m}\cdot\text{K}}\right)$
- k_{ins} - thermal conductivity of a insulation, $\left(\frac{\text{BTU}}{\text{ft}\cdot\text{h}\cdot\text{F}^\circ}, \frac{\text{W}}{\text{m}\cdot\text{K}}\right)$

3.2 General solution

We will start with analysing finite step analysis of fluid temperature drop over finite length change. Using graph for this would be the simplest approach.

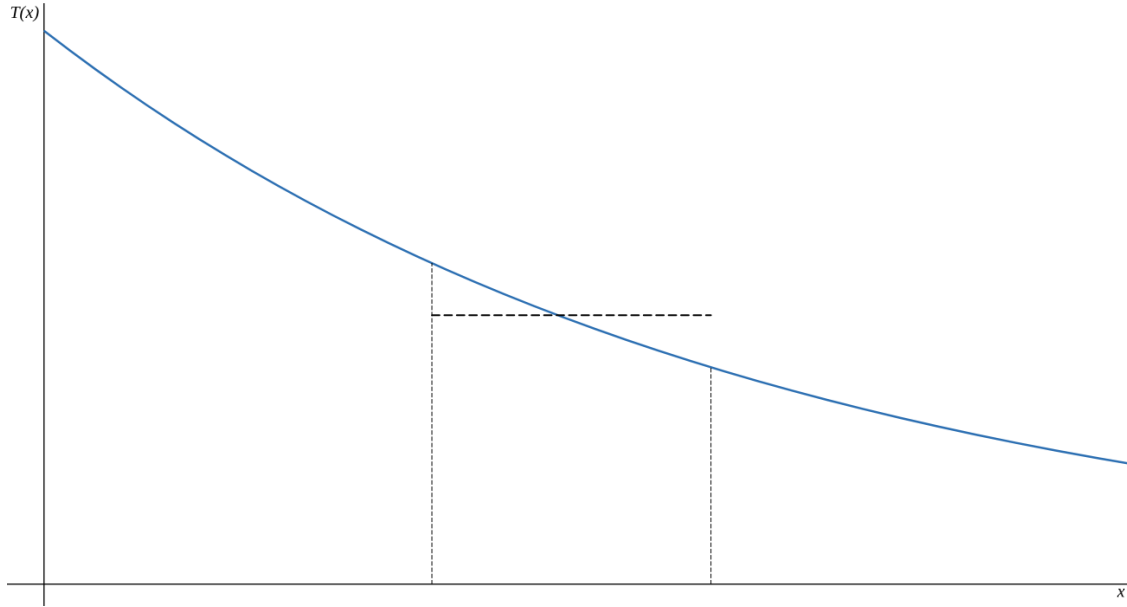


Figure 1: $\mathbf{T(x)}$ Function of $T(x)$

In the figure above we consider a change in temperature over a finite change in length on interval $[x_i, x_{i+1}]$, where temperature drops from $T_f(x_i)$ to $T_f(x_{i+1})$. The heat loss over piping segment $x_{i+1} - x_i = \Delta x_i$ is not constant, but drops with temperature. Because of that we consider some average temperature difference to find the average heat loss. The temperature we will use to find the difference is shown on the figure above as a horizontal line between x_i and x_{i+1} . With $w_i \in [0, 1]$ we can write that the average heat loss is:

$$\begin{aligned}\dot{Q}_{avg}(x_i) &= 2\pi(x_{i+1} - x_i) \cdot K(T_s(x_i)_{avg}) \cdot (T_f(x_i) - T_a)_{avg} \\ &= 2\pi\Delta x_i \cdot K(T_s(x_i + w_i\Delta x_i)) \cdot (T_f(x_i + w_i\Delta x_i) - T_a)\end{aligned}$$

Solar flux fragment:

$$A_{t,i}\dot{q} = \pi D_t \Delta x_i \dot{q}$$

And energy balance:

$$A_{t,i}\dot{q} - \dot{Q}_{avg}(x_i) = \dot{m}C \cdot (T_f(x_{i+1}) - T_f(x_i)) = \dot{m}C\Delta T_{f,i}$$

We connect above equations, rearrange it and drop the length dependency from temperatures for simplicity, so we can write:

$$\frac{1}{\frac{D_t\dot{q}}{2} - K(T_{s,i})\Delta T_{fa,i}} \frac{\Delta T_{f,i}}{\Delta x_i} = \frac{2\pi}{\dot{m}C} = \frac{1}{\lambda}$$

We can now take limit at $\Delta x_i \rightarrow 0$ and we will arrive at derivative form:

$$\frac{1}{\frac{D_t\dot{q}}{2} - K(T_s)\Delta T_{fa}} \frac{dT_f}{dx} = \frac{1}{\lambda} \quad (4)$$

And this is basically the equation we must integrate over. Here, the problem is that, K is a function of h and you can only write h in terms of T_s (methodology and correlations in chapter XX). Trying to integrate in fluid temperature domain would require constantly changing fluid temperature to surface temperature at each step of integration, which would add a lot of computation for basically no reason. Because of that, it is much simpler to carry the integration in surface temperature domain.

I'm organizing this blog in the order I came up with all of the solutions, and so we will start with $\dot{q} = 0$ solution.

3.2.1 No solar radiation solution

We just let $\dot{q} = 0$ and $dT_f(x) = dT_f$ for simplicity, then the equation simplifies to:

$$\frac{1}{K(T_s)\Delta T_{fa}} \frac{dT_f}{dx} = -\frac{1}{\lambda} \quad (5)$$

By utilizing thermal resistance network approach we can show that fluid and surface temperature are related by:

$$\frac{R_{fa}}{R_{sa}} \Delta T_{sa} = \Delta T_{fa} \quad (6)$$

Now we expand this thermal resistance fraction as:

$$\frac{R_{fa}}{R_{sa}} = \frac{D_th(T_s)}{24K(T_s)} = F(T_s) \quad (7)$$

We take natural logarithm on both sides of (6) and find:

$$d\ln F(T_s) + d\ln \Delta T_{sa} = d\ln \Delta T_{fa}$$

We express $d\ln F(T_s)$ in its differential form by the chain rule and take $d\ln \Delta T_{sa}$ out, obtaining:

$$\left(1 + \frac{d\ln F(T_s)}{d\ln \Delta T_{sa}}\right) d\ln \Delta T_{sa} = d\ln \Delta T_{fa}$$

Now, we rearrange a bit to fit (5) and arrive at differential relation connecting surface temperature and fluid temperature.

$$\left(\frac{1}{\Delta T_{sa}} + \frac{d \ln F(T_s)}{dT_s} \right) dT_s = \frac{dT_f}{\Delta T_{fa}} \quad (8)$$

All of this allows us to rewrite (5) in form ready to integrate:

$$\frac{1}{K(T_s)} \left(\frac{1}{\Delta T_{sa}} + \frac{d \ln F(T_s)}{dT_s} \right) \frac{dT_s}{dx} = -\frac{1}{\lambda} \quad (9)$$

And so we integrate:

$$\int_{T_{s,0}}^{T_s} \frac{1}{K(T)} \left(\frac{1}{T - T_a} + \frac{d \ln F(T)}{dT} \right) dT = -\frac{\Delta x}{\lambda} \quad (10)$$

And from now on, we will be referring to integrand part with no solar radiation simply as $I(T_s)$. I'm fairly certain that there is no closed form solution to this integral, and I have not even spend a second thinking how to find it. Instead, root-finding or minimization algorithms can be utilized to find surface temperature for given Δx . This will be discussed in chapter XX. Now we will discuss how to arrive at solution with non zero solar radiation term.

3.2.2 Solar radiation solution

Unfortunately, we cannot use results from 3.2.1 because thermal network approach changes under external heat flux and is not readily available. For that reason it is simpler to take energy balance of the surface area of piping at steady state:

$$P_t \dot{q} + 2\pi K(T_s) \Delta T_{fa} = P_t h(T_s) \Delta T_{sa} \quad (11)$$

Where P_t is a pipe perimeter. By introducing new parameter:

$$B(T_s) = \frac{\dot{q}}{h(T_s)} \quad (12)$$

and utilizing already introduced parameter $F(T_s)$, equation 11 can be rewritten as:

$$F(T_s)(\Delta T_{sa} - B(T_s)) = \Delta T_{fa} \quad (13)$$

Now we must take differential on both sides:

$$F(T_s) dT_s + \Delta T_{sa} dF(T_s) - F(T_s) dB(T_s) - B(T_s) dF(T_s) = dT_f$$

By the chain rule we can rewrite it as:

$$F(T_s) \Delta T_{sa} \left[\left(1 - \frac{dB(T_s)}{dT_s} \right) \frac{1}{\Delta T_{sa}} + \left(1 - \frac{B(T_s)}{\Delta T_{sa}} \right) \frac{d \ln F(T_s)}{dT_s} \right] dT_s = dT_f \quad (14)$$

Looking closely and letting $\dot{q} = 0$ we can see that (14) collapses to (8) by spotting that (13) must simplify to (6). Now, once we have differential relation between surface and fluid temperature we can go back to (4), where we plug $K(T_s) \Delta T_{fa}$ from (11) and (14) for dT_f .

Without rearranging and simplifying step by step we will just provide the final integral:

$$\int_{T_{s,0}}^{T_s} \frac{1}{K(T)} \left(1 - \frac{2B(T)}{T - T_a} \right)^{-1} \left[\left(1 - \frac{dB(T)}{dT} \right) \frac{1}{T - T_a} + \left(1 - \frac{B(T)}{T - T_a} \right) \frac{d \ln F(T)}{dT} \right] dT = -\frac{\Delta x}{\lambda} \quad (15)$$

Again, by letting $\dot{q} = 0$ we can see that (15) will collapse to (10) which is to be expected.